Closed-Form Tridiagonal Structure and $\mathcal{O}(n)$ Algorithms for the Linear and Nonlinear n-Mass Pendulum

Yonatan Ali Rodríguez Arias*[1] and Sandra Arellano González†[1]

[1]Instituto Politécnico Nacional, Escuela Superior de Ingeniería Mecánica y Eléctrica (ESIME) Unidad Zacatenco, Unidad Profesional "Adolfo López Mateos", Building 5, 3rd floor, Zacatenco, Alcaldía Gustavo A. Madero, 07738 Ciudad de México, Mexico

November 2026

Abstract

The linearized equations of the planar n-mass pendulum are organized by a cumulative mass-sum structure, $\sum_{i\geq\max(d,k)} m_i$, which we state in matrix form and prove directly from the kinetic energy. It induces the factorization $\mathbf{M} = \mathbf{L}\mathbf{E}^{\mathsf{T}}\mathbf{D}\mathbf{E}\mathbf{L}$ of the mass matrix, from which the main results follow: the inverse mass matrix and the dynamic matrix $\mathbf{M}^{-1}\mathbf{K}$ are tridiagonal in closed form, so the state-space operator is assembled in $\mathcal{O}(n)$ operations without solving any linear system; and the problem is similar to a symmetric Jacobi matrix, which proves that all natural frequencies are simple and that the k-th mode has exactly $k-1$ sign changes. The same structure yields a symplectic Störmer–Verlet integrator with $\mathcal{O}(n)$ cost per step and an $\mathcal{O}(n)$ energy evaluation. Numerically, the closed form is faster than the dense route by orders of magnitude and about four orders of magnitude more accurate at $n = 3000$; the spectrum of a uniform chain converges to that of the continuous hanging chain at rate $\mathcal{O}(n^{-1})$; and the integrator keeps a bounded energy error over 10^6 steps and remains practical at $n = 10^6$, where the dense transition matrix would require terabytes of storage. Beyond small oscillations, the rod tensions satisfy a tridiagonal system whose matrix is the angle-dependent continuation of the same structure; it gives the exact nonlinear accelerations in $\mathcal{O}(n)$ operations, about 4000 times faster than the dense formulation at $n = 3000$, and a symplectic RATTLE integrator with $\mathcal{O}(n)$ cost per step, validated in the chaotic regime. Code and data are openly available.

Keywords: n-mass pendulum; small oscillations; tridiagonal inverse; Jacobi matrix; symplectic integration; nonlinear dynamics; RATTLE; hanging chain

*Corresponding author: yonatanorion_84@hotmail.com; ORCID: 0000-0001-6080-2898

†ag.sandra012@gmail.com; ORCID: 0000-0002-2904-0474

1 Introduction

The pendulum remains one of the most fundamental systems in classical mechanics and continues to serve as a canonical model in multibody dynamics, robotics, vibration analysis, and control systems [1–6]. While the equations of motion for single and double pendulum systems are widely known, explicitly deriving the equations governing an arbitrary n-mass pendulum rapidly becomes cumbersome due to the combinatorial growth of inertial and trigonometric coupling terms.

1.1 Related Work and Framework Positioning

Several mature approaches exist for multibody equation generation and simulation. Featherstone's articulated-body algorithm [7] prioritizes recursive computational efficiency, achieving $\mathcal{O}(n)$ scaling suitable for real-time applications, though it operates implicitly through factorized spatial algebra. Kane's method [8] provides a highly systematic approach for deriving minimal-order equations but typically requires manual formulation for specific topologies. General recursive formulations covering serial and tree-topology systems are treated in [9, 10]. Furthermore, general-purpose symbolic environments (e.g., SymPy mechanics) can derive explicit equations but often yield massive, unstructured expressions for large n.

Closed-form equations of motion for the planar pendulum with n point masses have also been derived directly. Yesilyurt [11] obtained the general equations by Lagrangian mechanics and induction; when regrouped, the coefficient multiplying each pair of links is the cumulative mass sum used below. Charles [12] gave a general Lagrangian formulation together with a small-angle analysis. The classical theory of small vibrations of chain-like systems, including the oscillatory character of their spectra, is treated by Gantmacher and Krein [13]. We do not claim the cumulative mass sum itself as new; our aim is to state it in compact matrix form with a direct proof and to exploit it systematically in the assembly and solution of the linear problem.

The present work pursues a complementary objective: **explicit structural transparency**. Instead of focusing on raw computational speed for massive degrees of freedom, we make the cumulative mass-sum structure the organizing principle of the matrix assembly and of the linear solution strategies built on it.

The resulting symmetric positive-definite second-order system admits the standard generalized-eigenvalue (modal) solution. We use that solution as an independent reference and retain a state-space matrix-exponential implementation for applications in which a transition operator is convenient, such as control and repeated propagation. The matrix exponential is evaluated by a scaled-and-squared truncated Peano–Baker (Taylor) series; this numerical device is standard [14, 15] and is not claimed as a new matrix-exponential algorithm. The conceptual pipeline is illustrated in Figure 1.

The paper is organized as follows. Sections 2–4 develop the symbolic generation framework and introduce the cumulative mass-sum rule. Sections 5–6 derive the closed-form tridiagonal structure, the state-space system, the operator-based propagator and an $\mathcal{O}(n)$ structure-preserving integrator. Section 7 presents validation results in the linear regime. Section 8 extends the structure to the exact nonlinear dynamics, with an $\mathcal{O}(n)$ tension formulation and a symplectic RATTLE integrator, and validates it in the chaotic regime. Finally, Section

Lagrangian Formulation $\longrightarrow$ Cumulative Mass-Sum Rule $\longrightarrow$ Matrix Assembly

$\longrightarrow$ State-Space Linearization $\longrightarrow$ Matrix Exponential Propagation

Figure 1: Conceptual pipeline of the proposed framework, bridging explicit structural generation and operator-based linear propagation.

9 concludes.

2 Lagrangian Formulation of the n-Mass Pendulum

2.1 Equations of Motion

The dynamics of the system are derived from the Euler-Lagrange equations

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_j}\right) - \frac{\partial L}{\partial q_j} = 0, \tag{1}$$

where the Lagrangian is defined as $L = T - U$, with T the kinetic energy and U the potential energy. For the n-mass pendulum,

$$T = \frac{1}{2}\sum_{i=1}^{n} m_i v_i^2, \qquad U = g\sum_{i=1}^{n} m_i y_i,$$

and $v_i^2 = \dot{x}_i^2 + \dot{y}_i^2$. The generalized coordinates are $q_j = \theta_j$, $j = 1, \ldots, n$.

2.2 Coordinate Convention

Each generalized coordinate θ_i is measured with respect to the downward vertical direction, as illustrated in Figure 2. The Cartesian coordinates of the i-th mass are therefore

$$x_i = \sum_{k=1}^{i} l_k \sin\theta_k, \tag{2}$$

$$y_i = -\sum_{k=1}^{i} l_k \cos\theta_k. \tag{3}$$

The origin is taken at the pivot, with the positive x-axis to the right and the positive y-axis upward. This convention ensures that when all $\theta_k = 0$ (vertical downward equilibrium), $y_i = -\sum_{k=1}^{i} l_k$ and $x_i = 0$.

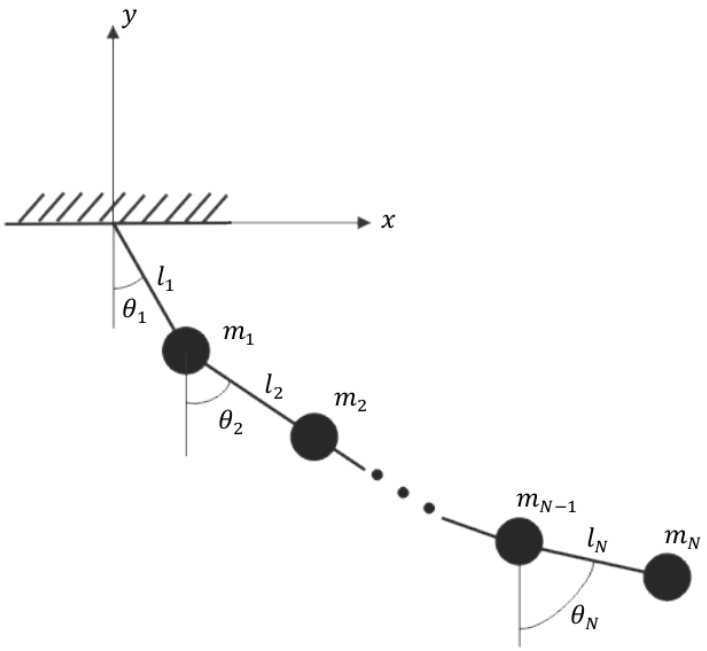


Figure 2: General n-mass pendulum with the absolute joint angle convention. Each angle θ_i is measured from the downward vertical. The coordinate origin is at the pivot.

2.3 Explicit Derivation of Kinetic Energy

Differentiating Eqs. (2) and (3) with respect to time gives the velocity components:

$$\dot{x}_i = \sum_{k=1}^{i} l_k \dot{\theta}_k \cos\theta_k, \tag{4}$$

$$\dot{y}_i = \sum_{k=1}^{i} l_k \dot{\theta}_k \sin\theta_k. \tag{5}$$

Hence,

$$v_i^2 = \dot{x}_i^2 + \dot{y}_i^2 = \sum_{k=1}^{i}\sum_{r=1}^{i} l_k l_r \dot{\theta}_k \dot{\theta}_r \big(\cos\theta_k \cos\theta_r + \sin\theta_k \sin\theta_r\big). \tag{6}$$

Using the trigonometric identity $\cos\theta_k \cos\theta_r + \sin\theta_k \sin\theta_r = \cos(\theta_k - \theta_r)$, we obtain the compact expression

$$v_i^2 = \sum_{k=1}^{i}\sum_{r=1}^{i} l_k l_r \dot{\theta}_k \dot{\theta}_r \cos(\theta_k - \theta_r). \tag{7}$$

Note that the double sum runs over all pairs of links up to the i-th mass. The total kinetic energy is then

$$T = \frac{1}{2}\sum_{i=1}^{n} m_i \sum_{k=1}^{i}\sum_{r=1}^{i} l_k l_r \dot{\theta}_k \dot{\theta}_r \cos(\theta_k - \theta_r). \tag{8}$$

2.4 Potential Energy and its Linearization

Substituting y_i from Eq. (3) gives the potential energy

$$U = g \sum_{i=1}^{n} m_i \left(- \sum_{k=1}^{i} l_k \cos\theta_k \right) = -g \sum_{k=1}^{n} \left(\sum_{i=k}^{n} m_i \right) l_k \cos\theta_k. \tag{9}$$

For small oscillations we expand $\cos\theta_k \approx 1 - \theta_k^2/2$; the constant term does not affect the dynamics, so the relevant part is

$$U_{\text{quad}} = \frac{g}{2} \sum_{k=1}^{n} \left(\sum_{i=k}^{n} m_i \right) l_k \theta_k^2. \tag{10}$$

No cross terms $\theta_i \theta_j$ $(i \neq j)$ appear because the expansion of each cosine depends only on its own angle. This crucial simplification leads to a diagonal stiffness matrix, as shown in Section 5.3.

3 Structural Pattern of the Equations of Motion

For small systems, the equations of motion can be derived manually. However, the combinatorial growth of velocity products makes manual derivation impractical for large n. The equations nevertheless share an invariant structural property (cf. [11]): every term in the d-th equation that involves coordinate θ_k or its derivatives contains the factor

$$\boxed{M_{d,k}^{\text{coeff}} = \sum_{i=\max(d,k)}^{n} m_i}. \tag{11}$$

3.1 The Cumulative Mass-Sum Rule as an Organizing Principle

This property follows directly from the kinetic energy.

Proposition 1. *For the planar n-mass pendulum with absolute angles,*

$$T = \tfrac{1}{2} \sum_{d=1}^{n} \sum_{k=1}^{n} S_{\max(d,k)}\, l_d l_k \cos(\theta_d - \theta_k)\, \dot\theta_d \dot\theta_k, \qquad S_j = \sum_{i=j}^{n} m_i.$$

Proof. In Eq. (7) mass m_i contributes to the pair (d,k) if and only if $d \le i$ and $k \le i$, i.e. $i \ge \max(d,k)$. Exchanging the order of summation in $T = \frac{1}{2}\sum_i m_i v_i^2$ collects these contributions into $S_{\max(d,k)}$. □

Applying the Euler–Lagrange equations (1) to this quadratic form, together with the potential energy of Section 2, yields the coefficients listed in Section 4; the velocity-squared terms arise from differentiating $\cos(\theta_d - \theta_k)$ and therefore carry the same factor $S_{\max(d,k)}$, and the gravitational term on joint d carries $S_d = S_{\max(d,d)}$.

This quantity represents the effective inertia dynamically shared by joints d and k. It acts as an organizing principle because the identical cumulative structure governs:

- diagonal and coupled acceleration terms,
- centrifugal and Coriolis-like velocity-squared terms,
- localized gravitational contributions.

Physical interpretation: when $k \geq d$, joint d contributes to the acceleration of all masses extending from coordinate k to the end of the chain. Conversely, when $k < d$, the inertial coupling relies on the deeper segment of the chain.

3.2 Explicit Example for $n = 4$

Applying the rule to the four-mass pendulum yields the first equation ($d = 1$), noting the proper dimensionality factor l_1:

$$\begin{aligned}
&(m_1 + m_2 + m_3 + m_4) l_1^2 \ddot{\theta}_1 + (m_2 + m_3 + m_4) l_1 l_2 \ddot{\theta}_2 \cos(\theta_1 - \theta_2) \\
&+ (m_3 + m_4) l_1 l_3 \ddot{\theta}_3 \cos(\theta_1 - \theta_3) + m_4 l_1 l_4 \ddot{\theta}_4 \cos(\theta_1 - \theta_4) \\
&+ (m_2 + m_3 + m_4) l_1 l_2 \dot{\theta}_2^2 \sin(\theta_1 - \theta_2) + (m_3 + m_4) l_1 l_3 \dot{\theta}_3^2 \sin(\theta_1 - \theta_3) \\
&+ m_4 l_1 l_4 \dot{\theta}_4^2 \sin(\theta_1 - \theta_4) + (m_1 + m_2 + m_3 + m_4) g l_1 \sin\theta_1 = 0
\end{aligned} \tag{12}$$

and the second equation ($d = 2$), scaled by l_2:

$$\begin{aligned}
&(m_2 + m_3 + m_4) l_2^2 \ddot{\theta}_2 + (m_3 + m_4) l_2 l_3 \ddot{\theta}_3 \cos(\theta_2 - \theta_3) + m_4 l_2 l_4 \ddot{\theta}_4 \cos(\theta_2 - \theta_4) \\
&+ (m_2 + m_3 + m_4) l_1 l_2 \ddot{\theta}_1 \cos(\theta_1 - \theta_2) + (m_3 + m_4) l_2 l_3 \dot{\theta}_3^2 \sin(\theta_2 - \theta_3) \\
&+ m_4 l_2 l_4 \dot{\theta}_4^2 \sin(\theta_2 - \theta_4) - (m_2 + m_3 + m_4) l_1 l_2 \dot{\theta}_1^2 \sin(\theta_1 - \theta_2) \\
&+ (m_2 + m_3 + m_4) g l_2 \sin\theta_2 = 0.
\end{aligned} \tag{13}$$

The third and fourth equations can be obtained using the same methodology. Every coefficient is exactly a cumulative mass sum: $\sum_{i=\max(d,k)}^{n} m_i$ multiplied by the product of lengths $l_d l_k$.

4 Algorithmic Generation Framework

The complete symbolic structure can be mapped into a concise algorithm using the mass-sum rule, eliminating complex parsers or manual recursion.

4.1 Matrix Assembly from the Cumulative Sums

We define the mass matrix $\mathbf{M}(\boldsymbol{\theta})$ and the Coriolis/centrifugal vector $\mathbf{c}(\boldsymbol{\theta}, \dot{\boldsymbol{\theta}})$ through the relation

$$\mathbf{M}(\boldsymbol{\theta})\ddot{\boldsymbol{\theta}} + \mathbf{c}(\boldsymbol{\theta}, \dot{\boldsymbol{\theta}}) + \mathbf{g}(\boldsymbol{\theta}) = \mathbf{0},$$

where $\mathbf{g}(\boldsymbol{\theta})$ is the gravitational torque vector. The cumulative mass-sum rule gives explicit formulas:

$$M_{jk}(\boldsymbol{\theta}) = \left(\sum_{i=\max(j,k)}^{n} m_i \right) l_j l_k \cos(\theta_j - \theta_k), \tag{14}$$

$$c_d(\boldsymbol{\theta}, \dot{\boldsymbol{\theta}}) = \sum_{k=1}^{n} \left(\sum_{i=\max(d,k)}^{n} m_i \right) l_d l_k \, \dot{\theta}_k^2 \sin(\theta_d - \theta_k), \tag{15}$$

$$g_d(\boldsymbol{\theta}) = \left(\sum_{i=d}^{n} m_i \right) g l_d \sin\theta_d. \tag{16}$$

The sine difference alone determines the sign of every velocity-squared term; no additional index-dependent sign is present. For example, when $d = 2$ and $k = 1$, $\sin(\theta_2 - \theta_1) = -\sin(\theta_1 - \theta_2)$, exactly as in Eq. (13). For the linearized case (small angles) we set $\cos(\cdot) = 1$, $\sin(\cdot) = 0$ except in the gravitational term where $\sin\theta_d \approx \theta_d$.

4.2 Algorithm for Linear System Assembly

The following algorithm constructs the linearized system matrices directly from the cumulative sums without symbolic differentiation.

Algorithm 1 Linear Matrix Assembly via Cumulative Mass-Sum Rule

Require: masses $m_1, \ldots, m_n$, lengths $l_1, \ldots, l_n$, g
Ensure: mass matrix $\mathbf{M}$, stiffness matrix $\mathbf{K}$

$S_n \leftarrow m_n$
for $j \leftarrow n-1$ **down to** 1 **do**
    $S_j \leftarrow m_j + S_{j+1}$
end for
for $j \leftarrow 1$ **to** n **do**
    for $k \leftarrow 1$ **to** n **do**
        $M_{jk} \leftarrow S_{\max(j,k)} l_j l_k$
    end for
    $K_{jj} \leftarrow S_j g l_j$
end for

4.3 Algorithmic Scaling

The suffix masses S_j require $\mathcal{O}(n)$ work and are computed once. The subsequent loop assigns n^2 entries of $\mathbf{M}$ and n nonzero entries of $\mathbf{K}$, so the assembly requires $\mathcal{O}(n^2)$ work and storage. Recomputing the sum $\sum_{i=\max(j,k)}^{n} m_i$ inside every matrix entry would instead require $\mathcal{O}(n^3)$ work and is deliberately avoided. The dense $\mathbf{M}$ is, however, not needed to form the state-space operator: Section 5.4 shows that $\mathbf{M}^{-1}\mathbf{K}$ is available in closed form in $\mathcal{O}(n)$ operations, whereas forming it by factorizing $\mathbf{M}$ costs $\mathcal{O}(n^3)$.

5 Linearization and Matrix Assembly

5.1 Small-Angle Approximation

For small oscillations, we assume $|\theta_j| \ll 1$ rad. The following approximations are applied:

$$\sin\theta_j \approx \theta_j, \quad \cos(\theta_i - \theta_j) \approx 1,$$

and velocity-squared terms (which are second order in small quantities) are neglected. Under these assumptions the equations of motion reduce to the linear system

$$\mathbf{M}\ddot{\boldsymbol{\theta}} + \mathbf{K}\boldsymbol{\theta} = \mathbf{0}, \tag{17}$$

where $\mathbf{M}$ and $\mathbf{K}$ are constant matrices.

5.2 Derivation of the Mass Matrix

Inserting the small-angle approximation $\cos(\theta_j - \theta_k) \approx 1$ into the general expression for $M_{jk}(\boldsymbol{\theta})$ yields the constant mass matrix

$$M_{jk} = \left(\sum_{i=\max(j,k)}^{n} m_i \right) l_j l_k. \tag{18}$$

Note that $M_{jk} = M_{kj}$ because $\max(j,k) = \max(k,j)$. Positive definiteness follows from a factorization: writing $\mathbf{L} = \operatorname{diag}(l_1, \ldots, l_n)$, $\mathbf{D} = \operatorname{diag}(m_1, \ldots, m_n)$ and $\mathbf{E}$ for the lower-triangular matrix of ones ($E_{ij} = 1$ for $i \geq j$), Proposition 1 gives

$$\mathbf{M} = \mathbf{L}\,\mathbf{E}^{\mathsf{T}}\mathbf{D}\,\mathbf{E}\,\mathbf{L}, \tag{19}$$

since $(\mathbf{E}^{\mathsf{T}}\mathbf{D}\mathbf{E})_{jk} = \sum_i m_i E_{ij} E_{ik} = S_{\max(j,k)}$. Because $\mathbf{D}$ is positive definite and $\mathbf{EL}$ is invertible for positive masses and lengths, $\mathbf{M}$ is symmetric positive definite.

5.3 Diagonal Structure of the Gravitational Stiffness Matrix

From the potential energy expansion (Eq. (10)), the gravitational torque on joint d is

$$g_d = \frac{\partial U_{\text{quad}}}{\partial \theta_d} = g \left(\sum_{i=d}^{n} m_i \right) l_d \theta_d.$$

Therefore the stiffness matrix $\mathbf{K}$ is diagonal:

$$K_{jk} = \left(\sum_{i=j}^{n} m_i \right) g l_j \, \delta_{jk}. \tag{20}$$

No off-diagonal entries appear because the quadratic potential contains only θ_j^2 terms; the coordinate convention (angles measured from the downward vertical) eliminates $\theta_i \theta_j$ cross couplings.

Because $\mathbf{M}$ and $\mathbf{K}$ are symmetric positive definite for positive masses and lengths, the generalized eigenproblem admits an $\mathbf{M}$-orthonormal modal matrix $\mathbf{U}$:

$$\mathbf{KU} = \mathbf{MU}\boldsymbol{\Omega}^2, \qquad \mathbf{U}^\mathsf{T}\mathbf{MU} = \mathbf{I}, \qquad \mathbf{U}^\mathsf{T}\mathbf{KU} = \boldsymbol{\Omega}^2, \tag{21}$$

where $\boldsymbol{\Omega} = \mathrm{diag}(\omega_1, \ldots, \omega_n)$. Since both $\mathbf{M}$ and $\mathbf{K}$ are positive definite, all natural frequencies are real and strictly positive, so there are neither zero-frequency rigid-body modes nor unstable roots. Moreover, the frequencies are simple, $0 < \omega_1 < \omega_2 < \cdots < \omega_n$ (Theorem 2 below). With $\boldsymbol{\theta} = \mathbf{Uq}$, the system decouples into $\ddot{q}_j + \omega_j^2 q_j = 0$, with the exact solution

$$q_j(t) = q_j(0)\cos(\omega_j t) + \frac{\dot{q}_j(0)}{\omega_j}\sin(\omega_j t). \tag{22}$$

This is the natural direct solution for the unforced linear second-order problem and is used below as an independent numerical reference.

5.4 Closed-Form Inverse Mass Matrix and Tridiagonal Dynamic Matrix

Although $\mathbf{M}$ is dense, its inverse is not. Throughout, $1/m_0 := 0$ and entries not listed are zero.

Theorem 1 (Closed-form inverse mass matrix)**.** *For positive masses and lengths,* $\mathbf{M}^{-1}$ *is the symmetric tridiagonal matrix with*

$$(\mathbf{M}^{-1})_{jj} = \frac{1}{l_j^2}\left(\frac{1}{m_j} + \frac{1}{m_{j-1}}\right), \qquad (\mathbf{M}^{-1})_{j,j-1} = (\mathbf{M}^{-1})_{j-1,j} = -\frac{1}{m_{j-1}\, l_{j-1}\, l_j}. \tag{23}$$

Proof. By Eq. (19), $\mathbf{M}^{-1} = \mathbf{L}^{-1}\mathbf{E}^{-1}\mathbf{D}^{-1}\mathbf{E}^{-\mathsf{T}}\mathbf{L}^{-1}$. Since $\mathbf{E}$ forms cumulative sums, $\mathbf{E}^{-1}$ forms backward differences: it is lower bidiagonal with $(\mathbf{E}^{-1})_{jj} = 1$ and $(\mathbf{E}^{-1})_{j,j-1} = -1$. Hence $(\mathbf{E}^{-1}\mathbf{D}^{-1}\mathbf{E}^{-\mathsf{T}})_{jk} = \sum_i (\mathbf{E}^{-1})_{ji}(\mathbf{E}^{-1})_{ki}/m_i$ vanishes for $|j-k| > 1$, equals $1/m_j + 1/m_{j-1}$ for $k = j$ and $-1/m_{j-1}$ for $k = j-1$. Scaling by $\mathbf{L}^{-1}$ on both sides gives Eq. (23). □

Remark 1. The matrix $(S_{\max(j,k)})_{jk} = \mathbf{E}^\mathsf{T}\mathbf{DE}$ is a single-pair (Green's) matrix, and the tridiagonality of inverses of such matrices is classical [13, 16]. To our knowledge, its explicit consequences for the n-mass pendulum stated below have not been exploited in the pendulum literature, where the dense $\mathbf{M}$ is assembled and factorized [11, 12, 17].

Corollary 1 (Tridiagonal dynamic matrix)**.** *The dynamic matrix* $\mathbf{Q} = \mathbf{M}^{-1}\mathbf{K}$ *is tridiagonal with*

$$Q_{jj} = \frac{g\, S_j}{l_j}\left(\frac{1}{m_j} + \frac{1}{m_{j-1}}\right), \qquad Q_{j,j-1} = -\frac{g\, S_{j-1}}{m_{j-1}\, l_j}, \qquad Q_{j-1,j} = -\frac{g\, S_j}{m_{j-1}\, l_{j-1}}. \tag{24}$$

Consequently the state-space operator $\mathcal{A}$ *is assembled in* $\mathcal{O}(n)$ *operations and storage, and* $\mathcal{A}\mathbf{R}$ *costs* $\mathcal{O}(n)$*.*

Proof. Multiply Eq. (23) on the right by the diagonal matrix $\mathbf{K} = g\,\mathrm{diag}(S_j l_j)$. □

Corollary 1 has a transparent physical meaning: in the linearized equations solved for the accelerations, link j couples only to its neighbours $j \pm 1$, as expected from a Newtonian picture in which each mass interacts only through the two adjacent rods, whose equilibrium tensions are gS_j and gS_{j+1}. The dense coupling of $\mathbf{M}$ is an artifact of the absolute-angle Lagrangian description; it disappears upon inversion.

Theorem 2 (Spectral structure)**.** *Let* $\mathbf{K}^{1/2} = \operatorname{diag}\big(\sqrt{gS_j l_j}\big)$ *and* $\mathbf{J} = \mathbf{K}^{1/2}\mathbf{M}^{-1}\mathbf{K}^{1/2}$*. Then:*

(i) $\mathbf{J}$ *is symmetric tridiagonal with* $J_{jj} = Q_{jj}$ *and*

$$J_{j,j-1} = J_{j-1,j} = -\frac{g\sqrt{S_{j-1}S_j}}{m_{j-1}\sqrt{l_{j-1}l_j}} < 0; \tag{25}$$

(ii) $\mathbf{Ku} = \omega^2\mathbf{Mu}$ *if and only if* $\mathbf{Jw} = \omega^2\mathbf{w}$ *with* $\mathbf{u} = \mathbf{K}^{-1/2}\mathbf{w}$*;*

(iii) all natural frequencies are simple: $0 < \omega_1 < \omega_2 < \cdots < \omega_n$*;*

(iv) the k*-th mode shape* $\mathbf{u}^{(k)}$ *has exactly* $k-1$ *sign changes.*

Proof. (i) follows from Eq. (23) and the diagonal form of $\mathbf{K}^{1/2}$. (ii) $\mathbf{M}^{-1}\mathbf{Ku} = \omega^2\mathbf{u}$ is equivalent to $\mathbf{K}^{1/2}\mathbf{M}^{-1}\mathbf{K}^{1/2}(\mathbf{K}^{1/2}\mathbf{u}) = \omega^2\mathbf{K}^{1/2}\mathbf{u}$. (iii) $\mathbf{J}$ is congruent to $\mathbf{M}^{-1}$, hence positive definite. Because every off-diagonal entry is non-zero, the rows of $\mathbf{Jw} = \lambda\mathbf{w}$ form a three-term recurrence that determines $\mathbf{w}$ uniquely from w_1; every eigenspace is therefore one-dimensional, and since $\mathbf{J}$ is symmetric all eigenvalues are simple. (iv) For a Jacobi matrix with negative off-diagonal entries, the eigenvector of the k-th smallest eigenvalue has exactly $k-1$ sign changes [13, Ch. II]; multiplication by the positive diagonal matrix $\mathbf{K}^{-1/2}$ preserves signs. □

Remark 2 (Numerical aspects)*.* High-frequency modes are strongly localized: for random chains with $n \approx 20$–40 some components fall below 10^{-20} relative to the largest, so the sign pattern in (iv) cannot be resolved in double precision. We verified (iv) in 80-digit arithmetic for 849 modes of 60 random chains ($2 \le n \le 24$) with no exception. Eigenvalues computed from $\mathbf{J}$ with a tridiagonal solver are not affected by this localization.

Algorithm 2 $\mathcal{O}(n)$ assembly of the dynamic and Jacobi matrices

Require: masses $m_1, \ldots, m_n$, lengths $l_1, \ldots, l_n$, g
Ensure: tridiagonal $\mathbf{Q} = \mathbf{M}^{-1}\mathbf{K}$ and Jacobi matrix $\mathbf{J}$

```
S_n ← m_n; for j ← n − 1 down to 1: S_j ← m_j + S_{j+1}
for j ← 1 to n do
    Q_jj ← J_jj ← g S_j (1/m_j + 1/m_{j−1}) / l_j          ▷ 1/m_0 := 0
    if j > 1 then
        Q_{j,j−1} ← −g S_{j−1}/(m_{j−1} l_j);   Q_{j−1,j} ← −g S_j/(m_{j−1} l_{j−1})
        J_{j,j−1} ← J_{j−1,j} ← −g √(S_{j−1} S_j)/(m_{j−1} √(l_{j−1} l_j))
    end if
end for
```

All natural frequencies then follow from a symmetric tridiagonal eigensolver in $\mathcal{O}(n^2)$ operations, and any selected subset (e.g. the lowest p modes, by bisection) in $\mathcal{O}(np)$, compared with $\mathcal{O}(n^3)$ for the dense generalized eigenproblem.

5.5 State-Space Representation and Control Readiness

Define the state vector $\mathbf{R}(t) = (\boldsymbol{\theta}(t), \dot{\boldsymbol{\theta}}(t))^{\mathsf{T}} \in \mathbb{R}^{2n}$. The linearized system can then be written as

$$\dot{\mathbf{R}} = \mathcal{A}\mathbf{R}, \qquad \mathcal{A} = \begin{pmatrix} \mathbf{0} & \mathbf{I} \\ -\mathbf{M}^{-1}\mathbf{K} & \mathbf{0} \end{pmatrix}, \tag{26}$$

whose lower-left block is the tridiagonal matrix $-\mathbf{Q}$ of Corollary 1. The matrix $\mathcal{A}$ is constant and possesses purely imaginary conjugate eigenvalue pairs $\pm i\omega_j$, reflecting the undamped Hamiltonian structure of small oscillations. This first-order form is not required for the modal solution; it is retained because a reusable transition operator is directly compatible with modern state-space control frameworks, observer/Kalman filter design, and linear-quadratic regulation (LQR) where input matrices $\mathbf{B}_u$ are coupled to the equations of motion.

6 Operator-Based Dynamic Propagation

6.1 Peano-Baker Series for the Transition Matrix

Because $\mathcal{A}$ is constant, the state transition matrix from 0 to t is the matrix exponential $\Phi(t, 0) = e^{\mathcal{A}t}$.

For the autonomous system considered here, the Peano–Baker series is exactly the Taylor series of the matrix exponential; the distinction becomes relevant only for time-dependent operators, where the Peano–Baker and Magnus expansions differ [18, 19]. A direct truncated Taylor series can be inaccurate when $\|\mathcal{A}\Delta t\|$ is large [14], so the implementation uses scaling and squaring.

A convenient implementation defines $\mathbf{X} = \mathcal{A}\Delta t$ and chooses $s = \max\{0, \lceil \log_2(\|\mathbf{X}\|_\infty/\eta)\rceil\}$ with $\eta = 1/2$. For $\mathbf{B} = \mathbf{X}/2^s$, the approximation is

$$\Phi(\Delta t) \approx \left(\sum_{k=0}^{N} \frac{\mathbf{B}^k}{k!}\right)^{2^s}. \tag{27}$$

Scaling controls the Taylor remainder before repeated squaring. With $N = 20$, the tested systems agree with both `scipy.linalg.expm` and the modal solution to double-precision round-off. This implementation is included for transparency; robust library routines based on established matrix-exponential algorithms, such as scaling and squaring with Padé approximants [15], should be preferred outside the tested parameter range.

6.2 Propagation Algorithm and Reusability

Once $\Phi(\Delta t)$ is precomputed (using the scaled series), the linear propagation over a time step is simply

$$\mathbf{R}(t + \Delta t) = \Phi(\Delta t)\mathbf{R}(t).$$

This requires only matrix-vector multiplications, making it highly efficient for repeated time stepping. Importantly, because $\Phi(\Delta t)$ is constant for fixed system parameters and time step, it can be precomputed once and reused across many simulations (e.g., parameter sweeps,

Monte Carlo runs, or real-time control loops). This reusability is a key practical advantage of the operator-based formulation.

6.3 Structure-Preserving $\mathcal{O}(n)$ Time Stepping

The transition matrix $\Phi(\Delta t)$ is dense: each step costs $\mathcal{O}(n^2)$, its computation $\mathcal{O}(n^3)$, and its storage $8(2n)^2$ bytes. For large n the tridiagonal form of Corollary 1 offers a cheaper route. Writing the linear equations as $\ddot{\boldsymbol{\theta}} = -\mathbf{Q}\boldsymbol{\theta}$, the Störmer–Verlet (leapfrog) scheme reads

$$\dot{\boldsymbol{\theta}}^{k+1/2} = \dot{\boldsymbol{\theta}}^{k} - \tfrac{\Delta t}{2}\mathbf{Q}\boldsymbol{\theta}^{k}, \qquad \boldsymbol{\theta}^{k+1} = \boldsymbol{\theta}^{k} + \Delta t\, \dot{\boldsymbol{\theta}}^{k+1/2}, \qquad \dot{\boldsymbol{\theta}}^{k+1} = \dot{\boldsymbol{\theta}}^{k+1/2} - \tfrac{\Delta t}{2}\mathbf{Q}\boldsymbol{\theta}^{k+1}. \tag{28}$$

Each step requires one tridiagonal product, i.e. $\mathcal{O}(n)$ operations, with no precomputation beyond Algorithm 2. The scheme is second order, symplectic and time-reversible; for linear systems it is stable for $\Delta t\, \omega_n < 2$, and its energy error then remains bounded, without secular drift [20]. The largest frequency ω_n is obtained in $\mathcal{O}(n)$ operations by bisection on the Jacobi matrix $\mathbf{J}$.

Monitoring the energy also does not require the dense $\mathbf{M}$. By Eq. (19),

$$\tfrac{1}{2}\dot{\boldsymbol{\theta}}^{\mathsf{T}}\mathbf{M}\dot{\boldsymbol{\theta}} = \tfrac{1}{2}\sum_{i=1}^{n} m_i \Big(\sum_{k\le i} l_k \dot{\theta}_k\Big)^2, \tag{29}$$

which is the kinetic energy written with the linearized horizontal velocity $\sum_{k\le i} l_k\dot{\theta}_k$ of mass i, and is evaluated in $\mathcal{O}(n)$ operations by a cumulative sum; the potential energy $\frac{1}{2}\boldsymbol{\theta}^{\mathsf{T}}\mathbf{K}\boldsymbol{\theta}$ is diagonal.

Two limitations should be stated. First, Verlet is not exact: its phase and energy errors are $\mathcal{O}(\Delta t^2)$, whereas $\Phi(\Delta t)$ is exact up to round-off. Second, the stability bound ties Δt to ω_n, which grows as a uniform chain is refined, so the total cost of simulating a fixed time interval grows faster than linearly in n. The two methods are therefore complementary: Φ for small and moderate n when maximal accuracy is required, Verlet when n is too large for dense matrices.

7 Numerical Validation (Linear Regime)

7.1 Linear Validation

To ensure a comprehensive evaluation and guarantee reproducibility, the linear framework was validated across four distinct test cases. In all scenarios, the gravitational acceleration was set to $g = 9.81$ m/s^2 and the systems were released from rest ($\dot{\boldsymbol{\theta}}(0) = \mathbf{0}$). The specific parameter sets–masses m (in kg), lengths l (in m), and initial angular displacements $\boldsymbol{\theta}(0)$ (in radians)–are defined as follows:

- $n = 2$ **(uniform):** $m_i = 1.0$, $l_i = 1.0$, and $\boldsymbol{\theta}(0) = [0.1, 0.05]^{\mathsf{T}}$.
- $n = 3$ **(uniform):** $m_i = 1.0$, $l_i = 1.0$, and $\boldsymbol{\theta}(0) = [0.1, 0.05, 0.02]^{\mathsf{T}}$.

- $n = 3$ **(non-uniform):** $m = [2.0, 1.5, 0.8]^\mathsf{T}$, $l = [1.2, 0.9, 0.6]^\mathsf{T}$, and $\boldsymbol{\theta}(0) = [0.1, 0.05, 0.02]^\mathsf{T}$.
- $n = 5$ **(uniform):** $m_i = 1.0$, $l_i = 1.0$, and $\boldsymbol{\theta}(0) = [0.1, 0.05, 0.02, 0.01, 0.005]^\mathsf{T}$.

These configurations provide a baseline for verifying the assembly and propagation across varying dimensionalities and inertial distributions. The scaled Peano–Baker series was truncated at $N = 20$ with a time step of $\Delta t = 0.01$ s. Reference solutions were generated both by direct matrix exponentiation (`scipy.linalg.expm`) and by the generalized-eigenvalue solution in Eq. (22).

Table 1 confirms that both independent solution paths agree with the reference matrix exponential at the level of double-precision round-off. The individual digits at this level depend on the platform and BLAS implementation; independent reruns give errors of the same order (below 2×10^{-14}).

Table 1: Maximum absolute error in $\theta_1(t)$ over $t \in [0, 8]$ s (reference: `scipy.linalg.expm`).

System	**Parameters**	Δt (s)	**Scaled series**	**Modal solution**
$n = 2$	uniform	0.01	6.88×10^{-15}	9.71×10^{-16}
$n = 3$	uniform	0.01	2.75×10^{-15}	1.57×10^{-15}
$n = 3$	non-uniform	0.01	1.60×10^{-15}	8.19×10^{-16}
$n = 5$	uniform	0.01	9.28×10^{-15}	1.31×10^{-15}

7.2 Closed-Form Structure: Cost, Accuracy and the Continuum Limit

Figure 4 and Table 2 compare the closed-form tridiagonal route with the dense route on chains with random masses $m_j \in [0.5, 2]$ and lengths $l_j \in [0.5, 2]/n$. The dense route forms $\mathbf{M}$ and $\mathbf{K}$ with vectorized operations, obtains $\mathbf{M}^{-1}\mathbf{K}$ by an LU solve and the frequencies with the generalized symmetric eigensolver `scipy.linalg.eigh(K, M)`; the tridiagonal route uses Eq. (24) and `scipy.linalg.eigh_tridiagonal` applied to $\mathbf{J}$. Timings are best-of-several wall times on a single thread of an Intel Xeon processor at 2.3 GHz (NumPy 2.5); absolute values are platform dependent, but the scaling is not. Accuracy is measured by the largest relative eigenpair residual $\|\mathbf{Q}\mathbf{u}_i - \omega_i^2\mathbf{u}_i\|/(\omega_i^2\|\mathbf{u}_i\|)$, evaluated with the exact $\mathcal{O}(n)$ operator of Eq. (24).

Forming $\mathbf{Q}$ takes about 10^{-5}–10^{-3} s up to $n = 10^5$, whereas the dense solve already takes 1.3 s at $n = 3000$, a factor of about 3.5×10^4. The full spectrum is about 50 times faster at $n = 3000$ and remains affordable at $n = 3 \times 10^4$, where the dense eigenproblem is out of reach. The tridiagonal route is also more accurate: the dense residual grows with the condition number of $\mathbf{M}$ (Section 7) and reaches 10^{-4} at $n = 3000$, about four orders of magnitude above the tridiagonal one.

As an independent, non-circular check, a uniform chain of total mass m_{tot} and length L ($m_j = m_{\mathrm{tot}}/n$, $l_j = L/n$) is compared with the continuous hanging chain, whose natural frequencies are the classical values $\omega_k = \frac{1}{2} j_{0,k}\sqrt{g/L}$ [21], with $j_{0,k}$ the k-th zero of the Bessel

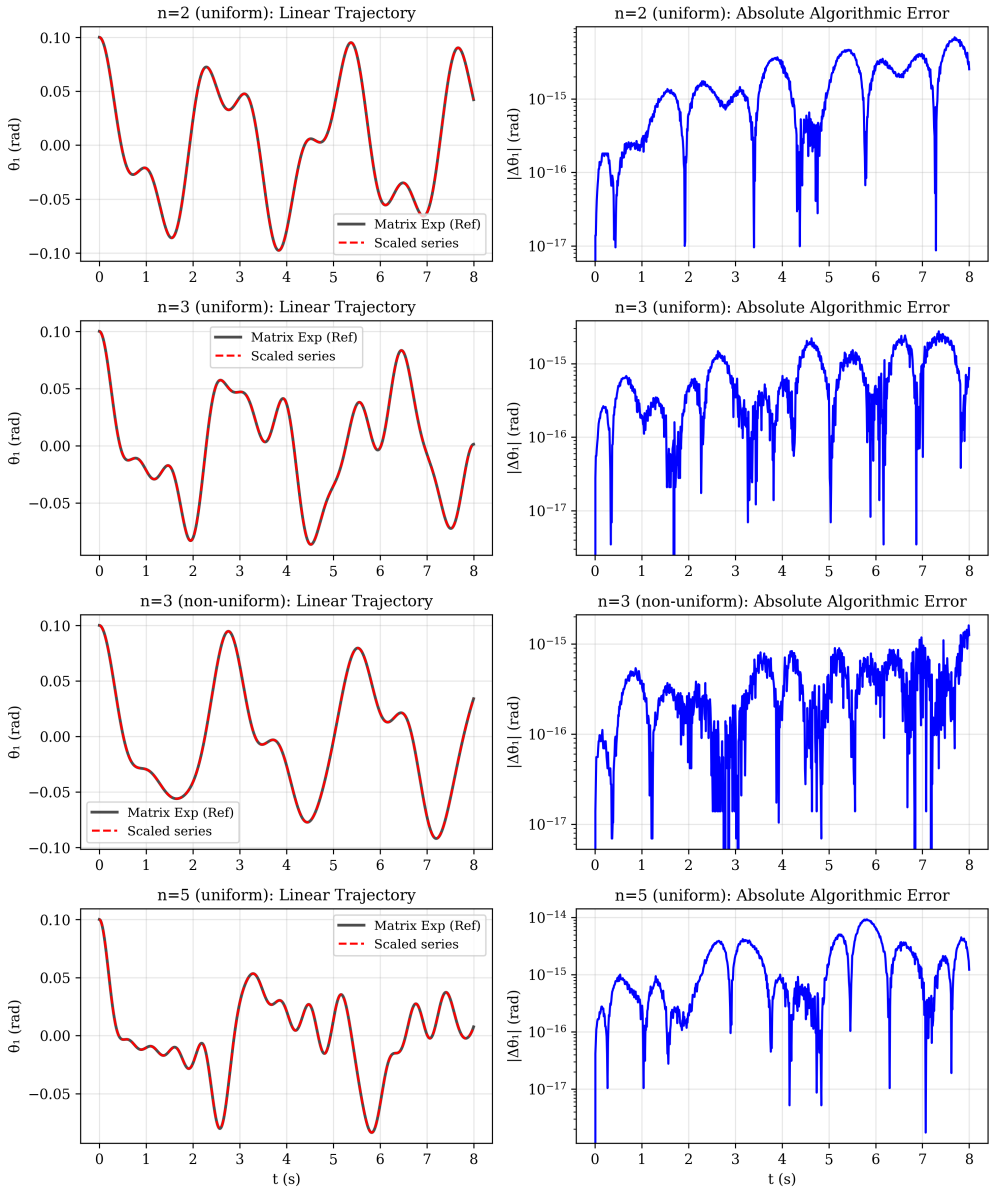


Figure 3: Linear validation results for the evaluated configurations. Left column: time-domain trajectories showing scaled-series stepping (dashed) versus the reference matrix exponential (solid). Right column: absolute algorithmic error confirming double-precision agreement.

function J_0. The three lowest frequencies converge at rate $\mathcal{O}(n^{-1})$ (Figure 4c); empirically the relative error is very close to $1/(4n)$ for large n (e.g. 3.05×10^{-5} at $n = 8192$).

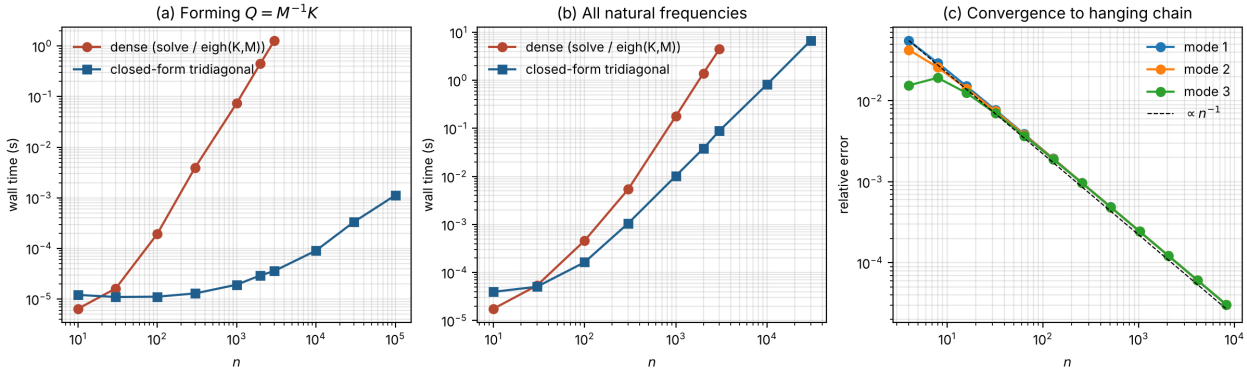


Figure 4: (a) Wall time to form $\mathbf{Q} = \mathbf{M}^{-1}\mathbf{K}$: dense LU solve versus the closed form of Corollary 1. (b) Wall time for all natural frequencies: dense generalized eigensolver versus the tridiagonal Jacobi form of Theorem 2. (c) Relative error of the three lowest frequencies of a uniform n-mass chain with respect to the continuous hanging chain.

Table 2: Cost (s) and largest relative eigenpair residual of the dense and closed-form tridiagonal routes (random chains, single thread).

	Forming $\mathbf{Q}$		All frequencies		Residual	
n	dense	closed form	dense	tridiagonal	dense	tridiagonal
100	1.9×10^{-4}	1.1×10^{-5}	4.6×10^{-4}	1.6×10^{-4}	9.2×10^{-10}	8.0×10^{-12}
1000	7.4×10^{-2}	1.9×10^{-5}	1.8×10^{-1}	1.0×10^{-2}	2.7×10^{-6}	1.4×10^{-9}
3000	1.3	3.6×10^{-5}	4.5	9.0×10^{-2}	1.1×10^{-4}	1.7×10^{-8}
30000	–	3.3×10^{-4}	–	6.7	–	–

7.3 Structure-Preserving $\mathcal{O}(n)$ Time Stepping

Figure 5 evaluates the integrator of Section 6.3. For the non-uniform $n = 3$ pendulum over 10^6 steps (panel a), the relative energy error of Verlet oscillates without growth: its maximum is 2.0×10^{-4} in the first tenth of the run and 2.4×10^{-4} in the last tenth for $\Delta t = 0.01\,\mathrm{s}$, and it decreases fourfold when Δt is halved. The transition matrix is far more accurate, but its round-off error accumulates linearly, from 1.1×10^{-12} to 1.2×10^{-11} over the same run. Against the exact modal solution, the Verlet error at $t = 8\,\mathrm{s}$ decreases as Δt^2 (panel b).

Panel (c) compares the cost per step on random chains, with $\Delta t = 0.5/\omega_n$. The Verlet step is cheaper than the dense matrix–vector product from $n \approx 100$; at $n = 2000$ it takes $8.6 \times 10^{-6}\,\mathrm{s}$ against $7.0 \times 10^{-3}\,\mathrm{s}$, a factor of about 800, to which the dense route adds 10.7 s to compute Φ and 128 MB to store it. Verlet remains practical at $n = 10^6$ ($9.4 \times 10^{-3}\,\mathrm{s}$ per step), where Φ would require about 32 TB.

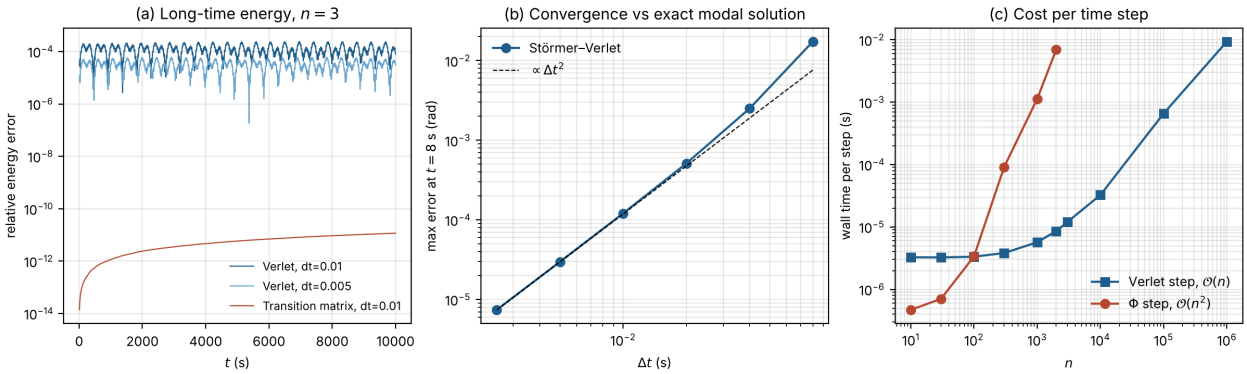


Figure 5: Störmer–Verlet on the closed-form tridiagonal system. (a) Relative energy error over 10^6 steps for the non-uniform $n = 3$ pendulum, compared with repeated application of the transition matrix. (b) Error at $t = 8\,\mathrm{s}$ against the exact modal solution. (c) Wall time per step on a single thread, Verlet versus the dense transition matrix.

7.4 Nonlinear Sign and Energy Check

Although the main numerical study is linear, the sign in the exact velocity vector was checked directly using the double-pendulum parameters $m_1 = m_2 = l_1 = l_2 = 1$, $g = 9.81$, $\theta_1(0) = 5°$, $\theta_2(0) = 0$, and zero initial velocities. The exact nonlinear equations were integrated over $0 \le t \le 6$ with the DOP853 Runge–Kutta method [22] using relative and absolute tolerances 10^{-12} and 10^{-14}, respectively. Table 3 shows the maximum drift in the physical energy $T + U$. With the velocity-squared term as derived, the drift remains at round-off. As a negative control, inserting a spurious index-dependent sign (a plausible transcription error that flips the term for $k < d$) produces a 10^{-4}-level drift, showing that the energy test is sensitive to this kind of error.

Table 3: Maximum absolute energy drift in the nonlinear double-pendulum sign check.

Velocity-squared term	$\max_t \lvert E(t) - E(0) \rvert$
As derived, $\sin(\theta_d - \theta_k)$	3.55×10^{-15}
Negative control (spurious sign for $k < d$)	2.26×10^{-4}

7.5 Consistency Check with RK4

For completeness, the state-space implementation was also checked against classical fourth-order Runge–Kutta (RK4) [22] on the same linear system. RK4 is a general-purpose time-stepper rather than a method tailored to this constant linear problem, so this comparison is only a discretization consistency check; the modal solution is the more appropriate direct baseline. Table 4 reports errors for a moderately large time step and for a ten-times-smaller RK4 step.

The scaled-series result demonstrates agreement with the matrix exponential after a fixed precomputation. It does not imply general superiority over Runge–Kutta methods, which are

Table 4: Consistency check for the $n = 3$ non-uniform pendulum (linear regime).

Method	Δt (s)	**Max error (rad)**
Scaled series ($N = 20$)	0.1	9.62×10^{-15}
RK4	0.1	9.19×10^{-4}
RK4	0.01	9.42×10^{-8}

designed for a broader class of nonlinear and nonautonomous problems.

8 Exact Nonlinear Dynamics in $\mathcal{O}(n)$

The tridiagonal structure is not restricted to small oscillations. Evaluating the exact accelerations from $\mathbf{M}(\boldsymbol{\theta})\ddot{\boldsymbol{\theta}} + \mathbf{c} + \mathbf{g} = \mathbf{0}$ by forming and factorizing the dense $\mathbf{M}(\boldsymbol{\theta})$ costs $\mathcal{O}(n^3)$ operations per evaluation. Recursive multibody algorithms reach $\mathcal{O}(n)$ for general serial chains [7, 9]; for a chain of point masses, a Newtonian formulation with the rod tensions as unknowns gives an explicit alternative that requires a single tridiagonal solve and connects directly to Theorem 1.

8.1 Tension Formulation

Let $\mathbf{e}_k = (\sin\theta_k, -\cos\theta_k)$ be the unit vector along link k, from mass $k-1$ to mass k (mass 0 is the pivot), $\mathbf{n}_k = (\cos\theta_k, \sin\theta_k)$ its normal, $\hat{\mathbf{y}}$ the upward unit vector, and T_k the tension of rod k, with $T_{n+1} := 0$. Newton's law for mass i and the kinematics of link k read

$$m_i\ddot{\mathbf{r}}_i = -T_i\mathbf{e}_i + T_{i+1}\mathbf{e}_{i+1} - m_i g\,\hat{\mathbf{y}}, \qquad \ddot{\mathbf{r}}_k - \ddot{\mathbf{r}}_{k-1} = l_k\ddot{\theta}_k\,\mathbf{n}_k - l_k\dot{\theta}_k^2\,\mathbf{e}_k, \tag{30}$$

with $\ddot{\mathbf{r}}_0 = \mathbf{0}$.

Theorem 3 (Exact nonlinear dynamics in $\mathcal{O}(n)$)**.** *For any state $(\boldsymbol{\theta}, \dot{\boldsymbol{\theta}})$, the tensions solve $\mathbf{B}(\boldsymbol{\theta})\mathbf{T} = \mathbf{r}$, where $\mathbf{B}(\boldsymbol{\theta})$ is symmetric positive definite and tridiagonal,*

$$B_{kk} = \frac{1}{m_k} + \frac{1}{m_{k-1}}, \qquad B_{k,k+1} = B_{k+1,k} = -\frac{\cos(\theta_{k+1} - \theta_k)}{m_k}, \qquad r_k = l_k\dot{\theta}_k^2 + g\cos\theta_1\,\delta_{k1}, \tag{31}$$

with $1/m_0 := 0$ and $\mathbf{B}(\mathbf{0}) = (\mathbf{E}^{\mathsf{T}}\mathbf{D}\mathbf{E})^{-1}$. The angular accelerations are then given explicitly by

$$l_k\ddot{\theta}_k = \frac{T_{k+1}\sin(\theta_{k+1} - \theta_k)}{m_k} - \frac{T_{k-1}\sin(\theta_k - \theta_{k-1})}{m_{k-1}} - g\sin\theta_1\,\delta_{k1}. \tag{32}$$

Both steps cost $\mathcal{O}(n)$ operations.

Proof. Dividing Eq. (30) for masses k and $k-1$ by their masses and subtracting (for $k = 1$, using $\ddot{\mathbf{r}}_0 = \mathbf{0}$) gives $l_k\ddot{\theta}_k\mathbf{n}_k - l_k\dot{\theta}_k^2\mathbf{e}_k = (-T_k\mathbf{e}_k + T_{k+1}\mathbf{e}_{k+1})/m_k - (-T_{k-1}\mathbf{e}_{k-1} + T_k\mathbf{e}_k)/m_{k-1} - g\hat{\mathbf{y}}\,\delta_{k1}$. Taking the scalar product with $\mathbf{e}_k$ and using $\mathbf{e}_j \cdot \mathbf{e}_k = \cos(\theta_j - \theta_k)$, $\mathbf{n}_k \cdot \mathbf{e}_k = 0$ and $\hat{\mathbf{y}} \cdot \mathbf{e}_1 = -\cos\theta_1$ yields Eq. (31); the product with $\mathbf{n}_k$, using $\mathbf{e}_j \cdot \mathbf{n}_k = \sin(\theta_j - \theta_k)$ and

$\hat{\mathbf{y}} \cdot \mathbf{n}_1 = \sin\theta_1$, yields Eq. (32). Moreover, $\mathbf{B} = \hat{\mathbf{G}}\boldsymbol{\mathcal{M}}^{-1}\hat{\mathbf{G}}^{\mathsf{T}}$, where $\boldsymbol{\mathcal{M}} = \operatorname{diag}(m_i\mathbf{I}_2)$ and row k of $\hat{\mathbf{G}} \in \mathbb{R}^{n\times 2n}$ contains $\mathbf{e}_k^{\mathsf{T}}$ in block k and $-\mathbf{e}_k^{\mathsf{T}}$ in block $k-1$. Since $\hat{\mathbf{G}}$ is block lower bidiagonal with non-zero diagonal blocks, it has full row rank and $\mathbf{B}$ is positive definite. At $\boldsymbol{\theta} = \mathbf{0}$ all cosines equal one and $\mathbf{B}(\mathbf{0}) = \mathbf{E}^{-1}\mathbf{D}^{-1}\mathbf{E}^{-\mathsf{T}}$, as in the proof of Theorem 1. □

Corollary 2 (Consistency with the linear theory)**.** *At rest, $T_k = gS_k$, the weight suspended below rod k. Linearizing Eq. (32) about rest gives $\ddot{\boldsymbol{\theta}} = -\mathbf{Q}\boldsymbol{\theta}$, with $\mathbf{Q}$ the tridiagonal matrix of Corollary 1.*

Proof. At rest $\mathbf{r} = g\,\mathbf{e}^{(1)}$, so $\mathbf{T} = g\,\mathbf{E}^{\mathsf{T}}\mathbf{D}\mathbf{E}\,\mathbf{e}^{(1)}$, whose k-th entry is $gS_{\max(k,1)} = gS_k$. Since $\dot{\theta}_k^2$ and $T_k - gS_k$ are of second order, to first order $l_k\ddot{\theta}_k = gS_{k+1}(\theta_{k+1} - \theta_k)/m_k - gS_{k-1}(\theta_k - \theta_{k-1})/m_{k-1} - g\theta_1\delta_{k1}$. With $S_{k+1} = S_k - m_k$ and $S_{k-1} = S_k + m_{k-1}$, the coefficient of θ_k becomes $-gS_k(1/m_k + 1/m_{k-1})/l_k = -Q_{kk}$, and those of $\theta_{k\pm1}$ coincide with $-Q_{k,k\pm1}$ in Eq. (24). □

Theorem 3 and Corollary 2 thus close the circle: the inverse mass matrix of the linear theory is the zero-angle value of the tension matrix of the nonlinear theory, and the physical interpretation of Corollary 1 (nearest-neighbour coupling through rod tensions) holds exactly at all amplitudes.

8.2 Structure-Preserving Nonlinear Integration

Any explicit integrator applied to $\dot{\mathbf{R}} = (\dot{\boldsymbol{\theta}}, \ddot{\boldsymbol{\theta}})$ inherits the $\mathcal{O}(n)$ cost of Theorem 3 per stage. For long simulations a structure-preserving alternative is RATTLE [23, 24], applied to the Cartesian positions $\mathbf{q}_i$ with constraints $\phi_k(\mathbf{q}) = \frac{1}{2}(|\mathbf{q}_k - \mathbf{q}_{k-1}|^2 - l_k^2) = 0$ and Lagrange multipliers proportional to the tensions:

$$\mathbf{q}^{k+1} = \mathbf{q}^k + \Delta t\,\mathbf{v}^k + \tfrac{\Delta t^2}{2}\boldsymbol{\mathcal{M}}^{-1}\big(\mathbf{F} - \mathbf{G}(\mathbf{q}^k)^{\mathsf{T}}\boldsymbol{\lambda}\big), \qquad \boldsymbol{\phi}(\mathbf{q}^{k+1}) = \mathbf{0}, \tag{33}$$

$$\mathbf{v}^{k+1} = \frac{\mathbf{q}^{k+1} - \mathbf{q}^k}{\Delta t} + \tfrac{\Delta t}{2}\boldsymbol{\mathcal{M}}^{-1}\big(\mathbf{F} - \mathbf{G}(\mathbf{q}^{k+1})^{\mathsf{T}}\boldsymbol{\mu}\big), \qquad \mathbf{G}(\mathbf{q}^{k+1})\,\mathbf{v}^{k+1} = \mathbf{0}, \tag{34}$$

where $\mathbf{G} = \partial\boldsymbol{\phi}/\partial\mathbf{q}$ and $\mathbf{F}$ is gravity. RATTLE is second order, time-reversible and symplectic on the constraint manifold, and it enforces the rod lengths to solver tolerance at every step. Because constraint k involves only masses $k-1$ and k, both the Newton matrix $\mathbf{G}(\mathbf{q}^{k+1})\boldsymbol{\mathcal{M}}^{-1}\mathbf{G}(\mathbf{q}^k)^{\mathsf{T}}$ for $\boldsymbol{\lambda}$ in Eq. (33) and the symmetric matrix $\mathbf{G}\boldsymbol{\mathcal{M}}^{-1}\mathbf{G}^{\mathsf{T}}$ for $\boldsymbol{\mu}$ in Eq. (34) are tridiagonal, with the same pattern as $\mathbf{B}(\boldsymbol{\theta})$. Each step therefore costs $\mathcal{O}(n)$ operations per Newton iteration; about two iterations per step were needed in our tests.

8.3 Numerical Results

Figure 6 summarizes the nonlinear results. (a) The tension formulation agrees with the dense formulation to a relative difference below 4×10^{-12} for $n \le 3000$; one evaluation takes 1.7×10^{-4} s at $n = 3000$ against 0.68 s for the dense route (about 4000 times faster), and 6.8×10^{-3} s at $n = 10^5$.

(b) For a strongly chaotic test, the non-uniform triple pendulum of Section 7 is released from the horizontal position ($\theta_i(0) = \pi/2$, $\dot{\theta}_i(0) = 0$) and integrated for 200 s; the energy

error is normalized by $m_{\mathrm{tot}} g L_{\mathrm{tot}}$. RATTLE with $\Delta t = 10^{-3}\,\mathrm{s}$ keeps the error bounded below 5.6×10^{-5} with no growth (3.1×10^{-5} in the first tenth of the run, 2.7×10^{-5} in the last). Classical RK4 with the same step is more accurate on this horizon but drifts, from 6×10^{-10} to 1.1×10^{-8}; adaptive DOP853 drifts from 3.7×10^{-6} to 9.6×10^{-5} at tolerance 10^{-6} and to 1.4×10^{-8} at 10^{-9}. We therefore do not claim that RATTLE is more accurate than high-order methods over moderate horizons; its advantages are the absence of secular drift, exact preservation of the constraints and symplecticity, which matter for long-time and statistical studies. All four methods use the $\mathcal{O}(n)$ formulation.

(c) In the same chaotic regime, the RATTLE error at $t = 2\,\mathrm{s}$ with respect to a DOP853 reference (tolerance 10^{-13}) decreases with observed order 2.00.

(d) As a cross-validation between the nonlinear and linear theories, a uniform $n = 50$ chain is released from its fundamental mode shape with amplitude ε. The maximum deviation from the linear modal solution over 5 s, relative to ε, decreases as ε^2 (observed order 1.99), as expected from the cubic leading nonlinearity.

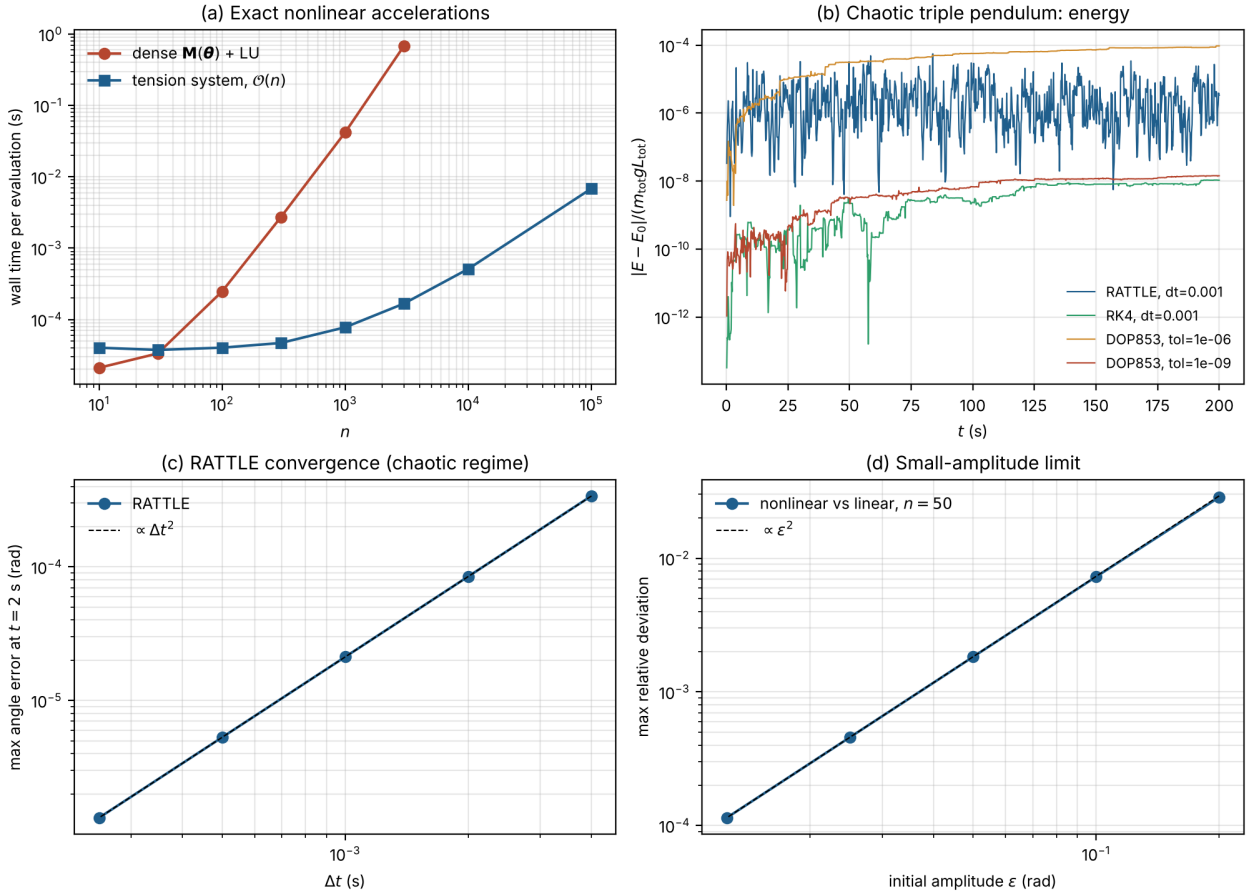


Figure 6: Exact nonlinear dynamics. (a) Wall time of one evaluation of the exact accelerations: dense $\mathbf{M}(\boldsymbol{\theta})$ with LU versus the tension formulation of Theorem 3. (b) Energy error for the chaotic triple pendulum released from the horizontal: RATTLE, RK4 and DOP853. (c) RATTLE convergence in the chaotic regime. (d) Deviation of the nonlinear solution of an $n = 50$ chain from the linear modal solution versus initial amplitude.

9 Conclusions

This work presented a structural framework for the linear and nonlinear dynamics of the planar n-mass pendulum. Its basis is the explicit matrix formulation, with a direct proof, of the cumulative mass-sum rule (Eq. 11) and its systematic use as an organizing principle, including the factorization (19) of the mass matrix. The factorization implies that the inverse mass matrix and the dynamic matrix $\mathbf{M}^{-1}\mathbf{K}$ are tridiagonal in closed form (Theorem 1, Corollary 1), so the state-space operator is assembled in $\mathcal{O}(n)$ operations without solving any linear system, and that the problem is similar to a Jacobi matrix, which proves that the natural frequencies are simple and that the k-th mode has $k-1$ sign changes (Theorem 2). Numerically, the closed form is faster than the dense route by orders of magnitude, more accurate by about four orders of magnitude at $n = 3000$, and reproduces the continuous hanging chain with $\mathcal{O}(n^{-1})$ convergence. The same structure yields a symplectic Störmer–Verlet integrator with $\mathcal{O}(n)$ cost per step and an $\mathcal{O}(n)$ energy evaluation, complementing the exact but dense transition matrix when n is large. The standard generalized-eigenvalue solution and a scaled-and-squared matrix-exponential implementation agree with `scipy.linalg.expm` to double-precision round-off in the tested cases. Because the transition matrix is constant, it can be precomputed once and reused across many simulations.

For the nonlinear dynamics, the rod tensions satisfy a symmetric positive-definite tridiagonal system whose zero-angle value is the core $(\mathbf{E}^{\mathsf{T}}\mathbf{D}\mathbf{E})^{-1}$ of the inverse mass matrix (Theorem 3), so the exact accelerations follow in $\mathcal{O}(n)$ operations and the linear theory is recovered as its small-amplitude limit (Corollary 2). The formulation is about 4000 times faster than the dense one at $n = 3000$, and a RATTLE integrator built on the same tridiagonal structure is second order with bounded energy error in the chaotic regime.

Natural extensions include rigid links with distributed mass, joint damping and actuation for control design, implicit or exponential integrators [25] for very fine chains, and experimental validation of the predicted spectra.

Funding

This research received no external funding.

Competing Interests

The authors declare no competing interests.

Data and Code Availability

The source code, validation and benchmark scripts and generated figures are openly available at `https://github.com/yonatanorion/Unified-n-Mass-Pendulum-Operator-Framework` and archived at Zenodo, doi: 10.5281/zenodo.20102179. All numerical results in this paper can be regenerated with `Code/validation/run_validation.py` and `Code/validation/benchmark_structure.py`, `benchmark_symplectic.py` and `benchmark_nonlinear.py`; the closed-form structure, the linear integrator and the nonlinear tension

formulation with RATTLE are implemented in `Code/src/structure.py`, `symplectic.py` and `chain.py`. The core implementation is reproduced in Appendix A.

A Core Python Implementation

```python
import numpy as np

def assemble_system(masses, lengths, g=9.81):
    """Assemble A, M, and K using precomputed suffix masses."""
    masses = np.asarray(masses, dtype=float)
    lengths = np.asarray(lengths, dtype=float)
    n = len(masses)
    S = np.cumsum(masses[::-1])[::-1]
    M = np.zeros((n, n))
    K = np.zeros((n, n))
    for j in range(n):
        for k in range(n):
            M[j, k] = S[max(j, k)] * lengths[j] * lengths[k]
        K[j, j] = S[j] * g * lengths[j]
    # Closed-form tridiagonal Q = M^{-1} K (Corollary 1), no linear solve.
    inv_m = 1.0 / masses
    inv_prev = np.concatenate(([0.0], inv_m[:-1]))
    Q = np.diag(g * S * (inv_m + inv_prev) / lengths)
    Q += np.diag(-g * S[:-1] * inv_m[:-1] / lengths[1:], -1)
    Q += np.diag(-g * S[1:] * inv_m[:-1] / lengths[:-1], 1)
    A = np.block([
        [np.zeros((n, n)), np.eye(n)],
        [-Q, np.zeros((n, n))]
    ])
    return A, M, K

def peano_baker_step(A, dt, N=20, eta=0.5):
    """Scaled-and-squared Taylor/Peano--Baker approximation."""
    B = A * dt
    norm_B = np.linalg.norm(B, ord=np.inf)
    s = 0 if norm_B <= eta else int(np.ceil(np.log2(norm_B / eta)))
    B = B / (2**s)
    dim = A.shape[0]
    Phi = np.eye(dim)
    term = np.eye(dim)
    for k in range(1, N + 1):
        term = (term @ B) / k
        Phi += term
    for _ in range(s):
        Phi = Phi @ Phi
    return Phi

def propagate_linear(Phi, R0, t):
    """Propagate linear system using precomputed transition matrix."""
    R = np.zeros((len(t), len(R0)))
    R[0] = R0
    for i in range(1, len(t)):
```

```
48            R[i] = Phi @ R[i-1]
49        return R
```

Listing 1: Core implementation of linear matrix assembly and propagation.

References

[1] L. G. Lobas. Generalized mathematical model of an inverted multilink pendulum with follower force. *International Applied Mechanics*, 41(5):566–572, 2005. doi: 10.1007/s10778-005-0125-1.

[2] Paweł Fritzkowski and Henryk Kamiński. Dynamics of a rope as a rigid multibody system. *Journal of Mechanics of Materials and Structures*, 3(6):1059–1075, 2008. doi: 10.2140/jomms.2008.3.1059.

[3] Paweł Fritzkowski and Henryk Kamiński. Dynamics of a rope modeled as a discrete system with extensible members. *Computational Mechanics*, 44(4):473–480, 2009. doi: 10.1007/s00466-009-0387-2.

[4] Waldemar Tomaszewski and Piotr Pieranski. Dynamics of ropes and chains: I. The fall of the folded chain. *New Journal of Physics*, 7:45, 2005. doi: 10.1088/1367-2630/7/1/045.

[5] S. Kawaji and K. Kanazawa. Control of double inverted pendulum with elastic joint. In *Proceedings IROS '91: IEEE/RSJ International Workshop on Intelligent Robots and Systems*, pages 946–951, 1991. doi: 10.1109/IROS.1991.174610.

[6] Raymond H. Plaut and Lawrence N. Virgin. Pendulum models of ponytail motion during walking and running. *Journal of Sound and Vibration*, 332(16):3768–3780, 2013. doi: 10.1016/j.jsv.2013.02.016.

[7] Roy Featherstone. *Rigid Body Dynamics Algorithms*. Springer, Boston, MA, 2008. doi: 10.1007/978-1-4899-7560-7.

[8] Thomas R. Kane and David A. Levinson. *Dynamics: Theory and Applications*. McGraw-Hill, New York, 1985.

[9] Abhinandan Jain. *Robot and Multibody Dynamics: Analysis and Algorithms*. Springer, Boston, MA, 2011. doi: 10.1007/978-1-4419-7267-5.

[10] Ahmed A. Shabana. *Dynamics of Multibody Systems*. Cambridge University Press, Cambridge, 4th edition, 2013. doi: 10.1017/CBO9781107337213.

[11] Boran Yesilyurt. Equations of motion formulation of a pendulum containing N-point masses, 2019. arXiv:1910.12610.

[12] Sergio Charles. The Lagrangian mechanics and pseudo-periodicity of the many-body planar pendulum system, 2019. arXiv:1911.04364.

[13] F. P. Gantmacher and M. G. Krein. *Oscillation Matrices and Kernels and Small Vibrations of Mechanical Systems*. American Mathematical Society, Providence, RI, revised edition, 2002. doi: 10.1090/chel/345.

[14] Cleve Moler and Charles Van Loan. Nineteen dubious ways to compute the exponential of a matrix, twenty-five years later. *SIAM Review*, 45(1):3–49, 2003. doi: 10.1137/S00361445024180.

[15] Nicholas J. Higham. The scaling and squaring method for the matrix exponential revisited. *SIAM Journal on Matrix Analysis and Applications*, 26(4):1179–1193, 2005. doi: 10.1137/04061101X.

[16] Gérard Meurant. A review on the inverse of symmetric tridiagonal and block tridiagonal matrices. *SIAM Journal on Matrix Analysis and Applications*, 13(3):707–728, 1992. doi: 10.1137/0613045.

[17] Aditya Bondada and Vishnu G. Nair. Dynamics of multiple pendulum system under a translating and tilting pivot. *Archive of Applied Mechanics*, 93(9):3699–3740, 2023. doi: 10.1007/s00419-023-02473-6.

[18] Wilhelm Magnus. On the exponential solution of differential equations for a linear operator. *Communications on Pure and Applied Mathematics*, 7(4):649–673, 1954. doi: 10.1002/cpa.3160070404.

[19] S. Blanes, F. Casas, J. A. Oteo, and J. Ros. The Magnus expansion and some of its applications. *Physics Reports*, 470(5–6):151–238, 2009. doi: 10.1016/j.physrep.2008.11.001.

[20] Ernst Hairer, Christian Lubich, and Gerhard Wanner. *Geometric Numerical Integration: Structure-Preserving Algorithms for Ordinary Differential Equations*, volume 31 of *Springer Series in Computational Mathematics*. Springer, Berlin, 2nd edition, 2006. doi: 10.1007/3-540-30666-8.

[21] Y. Verbin. Boundary conditions and modes of the vertically hanging chain. *European Journal of Physics*, 36(1):015005, 2015. doi: 10.1088/0143-0807/36/1/015005.

[22] Ernst Hairer, Syvert P. Nørsett, and Gerhard Wanner. *Solving Ordinary Differential Equations I: Nonstiff Problems*, volume 8 of *Springer Series in Computational Mathematics*. Springer, Berlin, 2nd edition, 1993. doi: 10.1007/978-3-540-78862-1.

[23] Hans C. Andersen. Rattle: A "velocity" version of the shake algorithm for molecular dynamics calculations. *Journal of Computational Physics*, 52(1):24–34, 1983. doi: 10.1016/0021-9991(83)90014-1.

[24] Benedict J. Leimkuhler and Robert D. Skeel. Symplectic numerical integrators in constrained Hamiltonian systems. *Journal of Computational Physics*, 112(1):117–125, 1994. doi: 10.1006/jcph.1994.1085.

[25] Marlis Hochbruck and Alexander Ostermann. Exponential integrators. *Acta Numerica*, 19:209–286, 2010. doi: 10.1017/S0962492910000048.